E-Commerce Analytics: Data Model (Power BI)

This repository documents the **data modeling** layer of an E-Commerce analytics project built in **Power BI**. The model is designed as a **Star Schema** to support fast, accurate analysis of sales, customers, products, countries, payments, and order status.

Overview

- **Tool:** Power BI Desktop (Model view)
- **Schema:** Star Schema
- **Structure:** 1 fact table + 6 dimension tables
- **Relationships:** all one-to-many (1:*), from each dimension to the fact table

Model view

> Add a screenshot of the Power BI Model view at `images/model-view.png`.

Model Diagram

```
erDiagram
    Dim_Customer ||--o{ Fact_Orders : "CustomerID"
    Dim_Date ||--o{ Fact_Orders : "Date_Id"
    Dim_Geography ||--o{ Fact_Orders : "Country_Id"
    Dim_OrderStatus ||--o{ Fact_Orders : "Status_Id"
    Dim_Payment ||--o{ Fact_Orders : "Payment_Id"
    Dim_Product ||--o{ Fact_Orders : "ProductID"
```

Fact Table: `Fact_Orders`

Stores the measurable business events (one row per order record).

Type	Columns
Keys	`Country_Id`, `CustomerID`, `Date_Id`, `Payment_Id`, `ProductID`, `Status_Id`, `SellerID`, `OrderID`

Type	Columns
Measures	`Quantity`, `UnitPrice`, `GrossAmount`, `Discount`, `DiscountAmount`, `NetSales`, `Tax`, `ShippingCost`, `TotalAmount`
Descriptive	`PriceBand`

Dimension Tables

Table	Columns	Business question answered
`Dim_Customer`	`CustomerID`, `CustomerType`, `OrderCount`	Who are our customers and what type are they?
`Dim_Date`	`Date_Id`, `OrderDate`	When were the orders placed?
`Dim_Geography`	`Country_Id`, `Country`	Which countries generate sales?
`Dim_OrderStatus`	`Status_Id`, `OrderStatus`, `StatusGroup`	What is the status of each order?
`Dim_Payment`	`Payment_Id`, `PaymentMethod`, `PaymentGroup`	How do customers pay?
`Dim_Product`	`ProductID`, `ProductName`, `Category`	What do we sell?

Relationships

From (dimension, "1" side)	To (fact, "*" side)
`Dim_Customer[CustomerID]`	`Fact_Orders[CustomerID]`
`Dim_Date[Date_Id]`	`Fact_Orders[Date_Id]`
`Dim_Geography[Country_Id]`	`Fact_Orders[Country_Id]`
`Dim_OrderStatus[Status_Id]`	`Fact_Orders[Status_Id]`

From (dimension, "1" side)	To (fact, "*" side)
`Dim_Payment[Payment_Id]`	`Fact_Orders[Payment_Id]`
`Dim_Product[ProductID]`	`Fact_Orders[ProductID]`

Filters flow from each dimension to the fact table, so selecting a country, category, or date range filters every measure consistently.

Why a Star Schema?

- **Performance:** simple joins and fast queries.
- **No redundancy:** customer and product attributes are stored once.
- **Ease of use:** straightforward DAX measures, slicers, and filters.
- **Scalability:** new dimensions can be added without restructuring the model.

Possible Next Steps

- Add a `Dim_Seller` dimension for `SellerID`.
- Build DAX measures on top of the model (total sales, average order value, discount rate).
- Develop dashboards for sales, customers, products, geography, payments, and order status.

Usage

1. Open the `.pbix` file in Power BI Desktop.
2. Go to **Model view** to inspect tables and relationships.
3. Use the model as the foundation for report pages and DAX measures.